MONGODB LIBRARY DATABASE PROJECT

Complete MongoDB Shell (mongosh) Operations

Database: libraryDB | Collection: books

BCA NoSQL / MongoDB Project

PROJECT INFORMATION

Database Name: libraryDB

Collection Name: books

Total initial records: 51 (Book ID 101-151)

Fields used: bookId, title, author, category, price, quantity, issued, year, publisher, status

Operations covered: Insert One, Insert Many, Find, Update One, Update Many, Delete One, Delete Many, Query Operations and Verification.

1. CREATE / SELECT DATABASE

Question: How do you create or select the database for the MongoDB library project?

```
use libraryDB
```

SCREENSHOT / OUTPUT

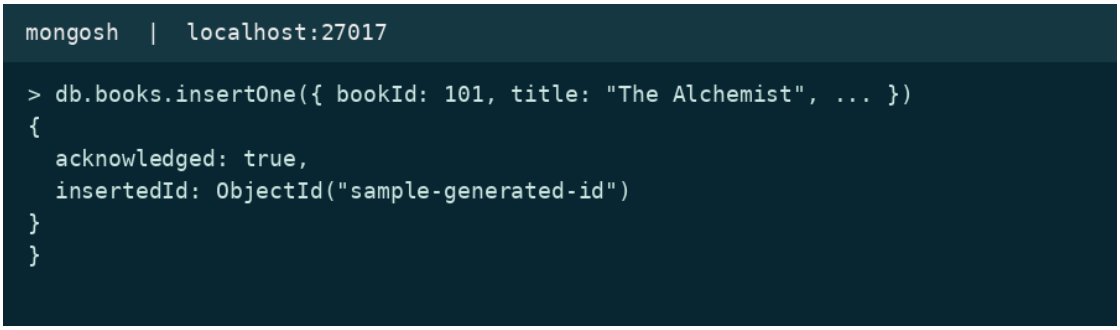


2. INSERT ONE

Question: Write a MongoDB command to insert one book record into the books collection.

```
db.books.insertOne({
  bookId: 101,
  title: "The Alchemist",
  author: "Paulo Coelho",
  category: "Fiction",
  price: 399,
  quantity: 7,
  issued: 2,
  year: 1988,
  publisher: "HarperCollins",
  status: "Available"
})
```

SCREENSHOT / OUTPUT



3. INSERT MANY

Question: Write a MongoDB command to insert multiple book records (Book ID 102-151) into the books collection.

```
db.books.insertMany([
{ bookId: 102, title: "To Kill a Mockingbird", author: "Harper Lee", category: "Fiction", price:
450, quantity: 6, issued: 3, year: 1960, publisher: "J.B. Lippincott", status: "Available" },
{ bookId: 103, title: "1984", author: "George Orwell", category: "Fiction", price: 320, quantity:
8, issued: 1, year: 1949, publisher: "Secker & Warburg", status: "Available" },
{ bookId: 104, title: "Pride and Prejudice", author: "Jane Austen", category: "Literature",
price: 299, quantity: 5, issued: 2, year: 1813, publisher: "T. Egerton", status: "Available" },
{ bookId: 105, title: "The Kite Runner", author: "Khaled Hosseini", category: "Fiction", price:
430, quantity: 4, issued: 2, year: 2003, publisher: "Riverhead Books", status: "Issued" },
{ bookId: 106, title: "A Brief History of Time", author: "Stephen Hawking", category: "Science",
price: 550, quantity: 5, issued: 2, year: 1988, publisher: "Bantam", status: "Available" },
{ bookId: 107, title: "Cosmos", author: "Carl Sagan", category: "Science", price: 620, quantity:
4, issued: 1, year: 1980, publisher: "Random House", status: "Available" },
{ bookId: 108, title: "Clean Code", author: "Robert C. Martin", category: "Technology", price:
699, quantity: 6, issued: 3, year: 2008, publisher: "Prentice Hall", status: "Available" },
{ bookId: 109, title: "The Pragmatic Programmer", author: "Andrew Hunt", category: "Technology",
price: 780, quantity: 5, issued: 2, year: 1999, publisher: "Addison-Wesley", status:
"Available" },
{ bookId: 110, title: "Introduction to Algorithms", author: "Thomas H. Cormen", category:
"Technology", price: 899, quantity: 3, issued: 2, year: 2009, publisher: "MIT Press", status:
"Issued" },
{ bookId: 111, title: "Atomic Habits", author: "James Clear", category: "Self Help", price: 499,
quantity: 9, issued: 4, year: 2018, publisher: "Avery", status: "Available" },
{ bookId: 112, title: "Deep Work", author: "Cal Newport", category: "Self Help", price: 399,
quantity: 7, issued: 2, year: 2016, publisher: "Grand Central", status: "Available" },
{ bookId: 113, title: "Wings of Fire", author: "A.P.J. Abdul Kalam", category: "Biography",
price: 350, quantity: 8, issued: 3, year: 1999, publisher: "Universities Press", status:
"Available" },
{ bookId: 114, title: "Long Walk to Freedom", author: "Nelson Mandela", category: "Biography",
price: 650, quantity: 4, issued: 1, year: 1994, publisher: "Little, Brown", status: "Issued" },
{ bookId: 115, title: "The Diary of a Young Girl", author: "Anne Frank", category: "Biography",
price: 299, quantity: 6, issued: 2, year: 1947, publisher: "Contact Publishing", status:
"Available" },
{ bookId: 116, title: "India After Gandhi", author: "Ramachandra Guha", category: "History",
price: 799, quantity: 3, issued: 1, year: 2007, publisher: "HarperCollins India", status:
"Available" },
{ bookId: 117, title: "Discovery of India", author: "Jawaharlal Nehru", category: "History",
price: 599, quantity: 5, issued: 2, year: 1946, publisher: "Signet", status: "Available" },
{ bookId: 118, title: "Sapiens", author: "Yuval Noah Harari", category: "History", price: 699,
quantity: 6, issued: 3, year: 2011, publisher: "Harper", status: "Issued" },
{ bookId: 119, title: "Gitanjali", author: "Rabindranath Tagore", category: "Literature", price:
249, quantity: 5, issued: 1, year: 1910, publisher: "Macmillan", status: "Available" },
{ bookId: 120, title: "The Great Gatsby", author: "F. Scott Fitzgerald", category: "Literature",
price: 350, quantity: 7, issued: 2, year: 1925, publisher: "Scribner", status: "Available" },
```

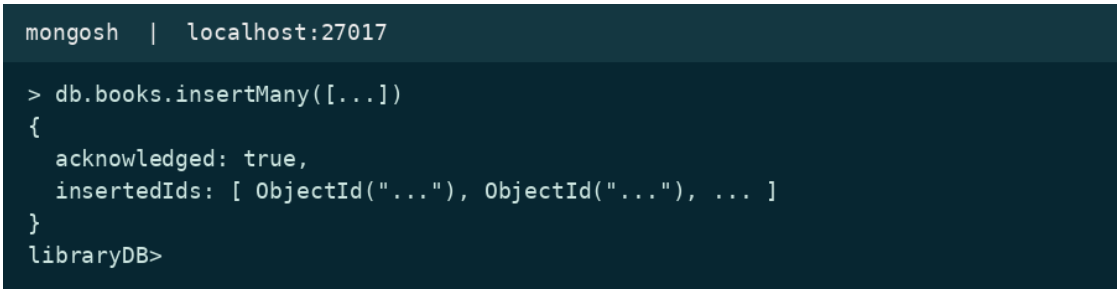
```
{ bookId: 121, title: "Hamlet", author: "William Shakespeare", category: "Literature", price:
275, quantity: 8, issued: 2, year: 1603, publisher: "Penguin Classics", status: "Available" },
{ bookId: 122, title: "The Odyssey", author: "Homer", category: "Literature", price: 450,
quantity: 4, issued: 1, year: 800, publisher: "Oxford University Press", status: "Available" },
{ bookId: 123, title: "One Hundred Years of Solitude", author: "Gabriel Garcia Marquez",
category: "Fiction", price: 520, quantity: 5, issued: 2, year: 1967, publisher: "Harper & Row",
status: "Available" },
{ bookId: 124, title: "The Book Thief", author: "Markus Zusak", category: "Fiction", price: 430,
quantity: 6, issued: 3, year: 2005, publisher: "Picador", status: "Issued" },
{ bookId: 125, title: "The Fault in Our Stars", author: "John Green", category: "Fiction", price:
380, quantity: 7, issued: 2, year: 2012, publisher: "Dutton", status: "Available" },
{ bookId: 126, title: "The Hobbit", author: "J.R.R. Tolkien", category: "Fiction", price: 399,
quantity: 6, issued: 2, year: 1937, publisher: "George Allen & Unwin", status: "Available" }
])
```

3. INSERT MANY (CONTINUED)

```
// Continue the same insertMany array with these records
{ bookId: 127, title: "Dune", author: "Frank Herbert", category: "Fiction", price: 499, quantity:
5, issued: 2, year: 1965, publisher: "Chilton Company", status: "Available" },
{ bookId: 128, title: "The Selfish Gene", author: "Richard Dawkins", category: "Science", price:
570, quantity: 4, issued: 1, year: 1976, publisher: "Oxford University Press", status:
"Available" },
{ bookId: 129, title: "Brief Answers to the Big Questions", author: "Stephen Hawking", category:
"Science", price: 520, quantity: 5, issued: 2, year: 2018, publisher: "Bantam", status:
"Available" },
{ bookId: 130, title: "Astrophysics for People in a Hurry", author: "Neil deGrasse Tyson",
category: "Science", price: 450, quantity: 6, issued: 2, year: 2017, publisher: "W. W. Norton",
status: "Issued" },
{ bookId: 131, title: "Python Crash Course", author: "Eric Matthes", category: "Technology",
price: 750, quantity: 7, issued: 3, year: 2015, publisher: "No Starch Press", status: "Available"
},
{ bookId: 132, title: "Head First Java", author: "Kathy Sierra", category: "Technology", price:
820, quantity: 4, issued: 2, year: 2005, publisher: "O"Reilly Media", status: "Available" },
{ bookId: 133, title: "Database System Concepts", author: "Abraham Silberschatz", category:
"Technology", price: 900, quantity: 3, issued: 2, year: 2019, publisher: "McGraw-Hill", status:
"Issued" },
{ bookId: 134, title: "Computer Networks", author: "Andrew S. Tanenbaum", category: "Technology",
price: 850, quantity: 5, issued: 2, year: 2011, publisher: "Pearson", status: "Available" },
{ bookId: 135, title: "Artificial Intelligence: A Modern Approach", author: "Stuart Russell",
category: "Technology", price: 999, quantity: 3, issued: 2, year: 2020, publisher: "Pearson",
status: "Available" },
{ bookId: 136, title: "The Story of My Experiments with Truth", author: "Mahatma Gandhi",
category: "Biography", price: 450, quantity: 5, issued: 2, year: 1927, publisher: "Navajivan
Publishing", status: "Available" },
{ bookId: 137, title: "Steve Jobs", author: "Walter Isaacson", category: "Biography", price: 699,
quantity: 4, issued: 1, year: 2011, publisher: "Simon & Schuster", status: "Available" },
{ bookId: 138, title: "Becoming", author: "Michelle Obama", category: "Biography", price: 750,
quantity: 5, issued: 2, year: 2018, publisher: "Crown", status: "Issued" },
{ bookId: 139, title: "Einstein: His Life and Universe", author: "Walter Isaacson", category:
"Biography", price: 620, quantity: 4, issued: 1, year: 2007, publisher: "Simon & Schuster",
status: "Available" },
{ bookId: 140, title: "A People’s History of the United States", author: "Howard Zinn", category:
"History", price: 680, quantity: 3, issued: 1, year: 1980, publisher: "Harper Perennial", status:
"Available" },
{ bookId: 141, title: "The Argumentative Indian", author: "Amartya Sen", category: "History",
price: 500, quantity: 5, issued: 2, year: 2005, publisher: "Farrar, Straus and Giroux", status:
"Available" },
{ bookId: 142, title: "The Wonder That Was India", author: "A.L. Basham", category: "History",
price: 590, quantity: 4, issued: 1, year: 1954, publisher: "Sidgwick & Jackson", status:
"Available" },
{ bookId: 143, title: "Man’s Search for Meaning", author: "Viktor E. Frankl", category: "Self
Help", price: 420, quantity: 7, issued: 3, year: 1946, publisher: "Beacon Press", status:
"Available" },
{ bookId: 144, title: "Think and Grow Rich", author: "Napoleon Hill", category: "Self Help",
price: 299, quantity: 8, issued: 3, year: 1937, publisher: "The Ralston Society", status:
"Available" },
{ bookId: 145, title: "How to Win Friends and Influence People", author: "Dale Carnegie",
category: "Self Help", price: 349, quantity: 7, issued: 2, year: 1936, publisher: "Simon &
Schuster", status: "Issued" },
{ bookId: 146, title: "The Power of Now", author: "Eckhart Tolle", category: "Self Help", price:
399, quantity: 6, issued: 2, year: 1997, publisher: "Namaste Publishing", status: "Available" },
{ bookId: 147, title: "Oxford Advanced Learner’s Dictionary", author: "Oxford University Press",
category: "Reference", price: 899, quantity: 5, issued: 1, year: 2020, publisher: "Oxford
University Press", status: "Available" },
{ bookId: 148, title: "Merriam-Webster Collegiate Dictionary", author: "Merriam-Webster",
category: "Reference", price: 799, quantity: 4, issued: 1, year: 2021, publisher: "Merriam-
```

```
Webster", status: "Available" },
{ bookId: 149, title: "The Chicago Manual of Style", author: "University of Chicago Press",
category: "Reference", price: 850, quantity: 3, issued: 1, year: 2017, publisher: "University of
Chicago Press", status: "Available" },
{ bookId: 150, title: "Guinness World Records 2024", author: "Guinness World Records", category:
"Reference", price: 1200, quantity: 2, issued: 1, year: 2023, publisher: "Guinness World
Records", status: "Available" },
{ bookId: 151, title: "Lost in the Library", author: "Mira Grant", category: "Fiction", price:
360, quantity: 1, issued: 0, year: 2022, publisher: "Orbit", status: "Lost" }
```

SCREENSHOT / OUTPUT



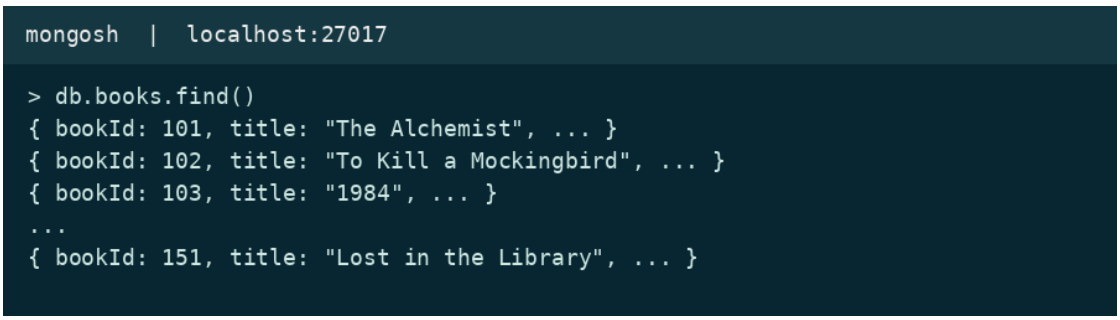
Note: Run Insert One first only when building the database from scratch. The Insert Many command starts from Book ID 102, so the final initial set contains 51 records (101-151).

4. FIND / DISPLAY ALL BOOKS

Question: How do you display all documents stored in the books collection?

```
db.books.find()
```

SCREENSHOT / OUTPUT

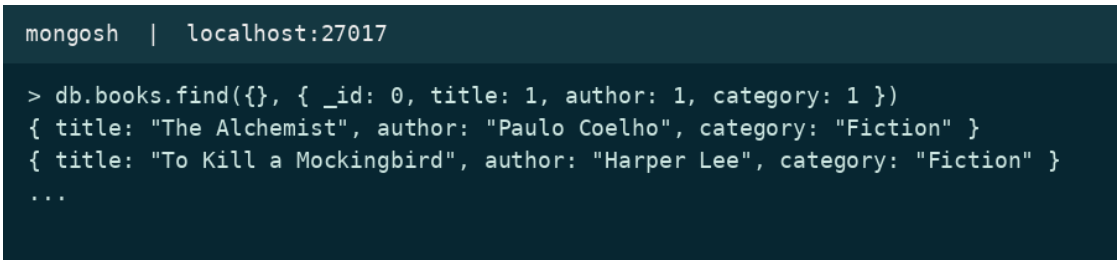


5. PROJECTION

Question: How do you display only the title, author and category fields?

```
db.books.find(
  {},
  { _id: 0, title: 1, author: 1, category: 1 }
)
```

SCREENSHOT / OUTPUT



6. UPDATE ONE

Question: How do you update the price and quantity of a particular book?

```
db.books.updateOne(
  { bookId: 108 },
  { $set: { price: 749, quantity: 7 } }
)
```

SCREENSHOT / OUTPUT



7. UPDATE MANY

Question: How do you increase the quantity of all Technology books by 1?

```
db.books.updateMany(
  { category: "Technology" },
  { $inc: { quantity: 1 } }
)
```

SCREENSHOT / OUTPUT



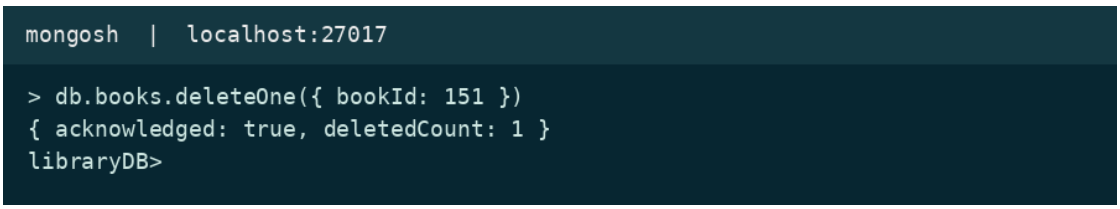
Why $inc? It increases the existing numeric quantity without replacing the whole document.

8. DELETE ONE

Question: How do you delete the book whose Book ID is 151?

```
db.books.deleteOne({ bookId: 151 })
```

SCREENSHOT / OUTPUT



9. DELETE MANY

Question: How do you delete all books whose status is Lost?

```
db.books.deleteMany({ status: "Lost" })
```

SCREENSHOT / OUTPUT



10. QUERY OPERATIONS - A. PRICE GREATER THAN 500

Question: How do you find all books whose price is greater than or equal to 500?

Concept: $gt / $gte

```
db.books.find({ price: { $gte: 500 } })
```

SCREENSHOT / OUTPUT

```
mongosh  |  localhost:27017

> db.books.find({ price: { $gte: 500 } })
...
sample query result shown in mongosh...
// Books matching condition: 27
```

10. QUERY OPERATIONS - B. AVAILABLE BOOKS

Question: How do you find all books that are currently available?

Concept: Equality filter

```
db.books.find({ status: "Available" })
```

SCREENSHOT / OUTPUT

```
mongosh  |  localhost:27017

> db.books.find({ status: "Available" })
...
sample query result shown in mongosh...
// Available records before delete operations: 41
```

10. QUERY OPERATIONS - C. FICTION BOOKS

Question: How do you find all books in the Fiction category?

Concept: Equality filter

```
db.books.find({ category: "Fiction" })
```

SCREENSHOT / OUTPUT

```
mongosh  |  localhost:27017

> db.books.find({ category: "Fiction" })
...
sample query result shown in mongosh...
// Fiction records: 10
```

10. QUERY OPERATIONS - D. BOOKS PUBLISHED AFTER 2010

Question: How do you find books published after the year 2010?

Concept: Comparison operator

```
db.books.find({ year: { $gt: 2010 } })
```

SCREENSHOT / OUTPUT

```
mongosh  |  localhost:27017

> db.books.find({ year: { $gt: 2010 } })
...
sample query result shown in mongosh...
// Initial matching records: 17
```

10. QUERY OPERATIONS - E. PRICE BETWEEN 300 AND 700

Question: How do you find books with price from 300 to 700?

Concept: Logical AND

```
db.books.find({ $and: [ { price: { $gte: 300 } }, { price: { $lte: 700 } } ] })
```

SCREENSHOT / OUTPUT

```
mongosh  |  localhost:27017

> db.books.find({ $and: [ { price: { $gte: 300 } }, { price: { $lte: 700 } } ] })
...
sample query result shown in mongosh...
// Initial matching records: 33
```

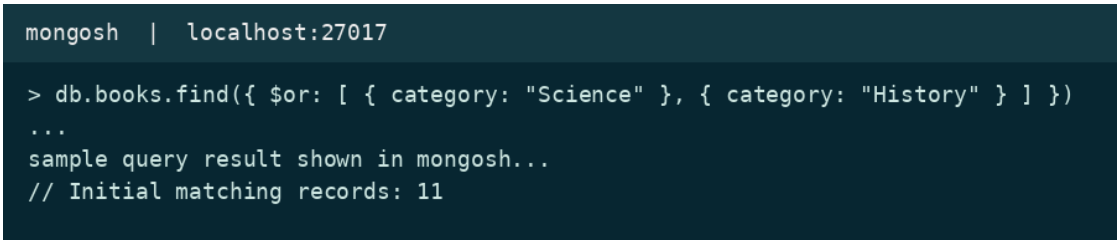
10. QUERY OPERATIONS - F. MULTIPLE CATEGORIES

Question: How do you find books that are either Science or History?

Concept: Logical OR

```
db.books.find({ $or: [ { category: "Science" }, { category: "History" } ] })
```

SCREENSHOT / OUTPUT



10. QUERY OPERATIONS - G. MOST EXPENSIVE BOOKS

Question: How do you display books sorted by price from highest to lowest?

Concept: Sort

```
db.books.find().sort({ price: -1 })
```

SCREENSHOT / OUTPUT

```
mongosh  |  localhost:27017

> db.books.find().sort({ price: -1 })
...
sample query result shown in mongosh...
// Highest initial price: Rs. 1200
```

10. QUERY OPERATIONS - H. LOWEST PRICED BOOKS

Question: How do you display books sorted by price from lowest to highest?

Concept: Sort

```
db.books.find().sort({ price: 1 })
```

SCREENSHOT / OUTPUT

```
mongosh  |  localhost:27017

> db.books.find().sort({ price: 1 })
...
sample query result shown in mongosh...
// Lowest initial price: Rs. 249
```

10. QUERY OPERATIONS - I. COUNT DOCUMENTS

Question: How do you count documents in the collection?

Concept: countDocuments

```
db.books.countDocuments({})
```

SCREENSHOT / OUTPUT

```
mongosh  |  localhost:27017

> db.books.countDocuments({})
...
sample query result shown in mongosh...
// Initial count: 51
```

10. QUERY OPERATIONS - J. TECHNOLOGY BOOK COUNT

Question: How do you count all Technology books?

Concept: countDocuments with filter

```
db.books.countDocuments({ category: "Technology" })
```

SCREENSHOT / OUTPUT

```
mongosh  |  localhost:27017

> db.books.countDocuments({ category: "Technology" })
...
sample query result shown in mongosh...
// Technology count: 8
```

11. VERIFICATION COMMANDS

Question: How do you verify the collection, indexes, count and sample records?

```
show collections
db.books.countDocuments({})
db.books.getIndexes()
db.books.findOne({ bookId: 101 })
```

SCREENSHOT / OUTPUT

```
mongosh  |  localhost:27017

> show collections
books
> db.books.countDocuments({})
51
> db.books.findOne({ bookId: 101 })
{ bookId: 101, title: "The Alchemist", ... }
```

12. CONCLUSION

The Library Management System demonstrates how MongoDB can store and manipulate library records using a document-oriented NoSQL model. The project covers database selection, document insertion, searching, projection, updating, deleting, sorting, counting and logical query operations.

IMPORTANT SUBMISSION NOTE

The dark terminal panels in this document are sample output illustrations, not screenshots from a live MongoDB installation. Before final submission, run the commands in mongosh or MongoDB Compass and replace the sample output panels with your own screenshots.

VIVA / PRACTICAL QUESTIONS

What is MongoDB and why is it called a NoSQL database?

What is the difference between a database and a collection in MongoDB?

What is a document?

What is the use of insertMany()?

What is the difference between updateOne() and updateMany()?

What do $gt, $gte, $lt and $lte mean?

What is the purpose of projection?

What is the difference between deleteOne() and deleteMany()?